STUDENT PROJECTS LABORATORY - LOCKOUT / TAGOUT [LOTO] PROCEDURE

ENERGY ISOLATION SPECIFICATION | ASSET: FAC-01 | TAG: FAC-SAFETY-01

Safety Hub, LOTO Lock Center, High-Voltage Battery Bench & Paint/Prep Booth

Authorized LOTO Personnel: Sr. Instructional Lab Supervisor (Max Wright) & Student Research Assistants (SRAs)

Notification Channel: Max Wright - Teams / Text Message | Zone: Zone 6: Facility & Infrastructure (FAC)

STEP 1 - NORMAL MACHINE SHUTDOWN PROCEDURE

Turn off Paint/Prep Booth exhaust fan switch; isolate high-voltage battery packs via manual service disconnect (MSD).

STEP 2 - HAZARDOUS ENERGY SOURCES, MAGNITUDES & ISOLATION POINTS

Source #1: Electrical / Chemical (HV Battery Bench) (>55-60V DC High-Voltage Tractive Packs & Li-Ion Cells)

Isolation Location: Dedicated High-Voltage Battery Bench & quarantine drum.

Lockout Procedure: Remove Manual Service Disconnect (MSD), apply insulated HV lockout cover, and verify 0V DC across main busbars with 1000V rated multimeter.

STEP 3 - STORED / RESIDUAL ENERGY DISSIPATION

Wait for HV bus discharge circuit to drop pack voltage < 5V DC; keep Class D & ABC extinguishers at arm’s reach.

STEP 4 - ZERO-ENERGY VERIFICATION (’TRY’ STEP)

Live-Dead-Live voltage verification using rated digital multimeter.